

Matthew Hounslow

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Work Experience

Software Developer

Oct. 2025 – Present

Sundae Labs

Remote

Shielded Technologies | Compact Language

- Test new features in Compact, Midnight's smart contract language, from compiler behaviour and language semantics through contract execution and integration with the wider network.
- Built multi-network end-to-end infrastructure that automates the full contract lifecycle and lets teams pin component versions or feature branches, allowing QA to test changes as they are developed and catch compatibility and state regressions before release.

Input Output (IOHK) | Acropolis Node

- Develop core ledger-state and indexing modules for Acropolis, a modular Cardano node in Rust whose independent components communicate through Cariatid's publish/subscribe message bus.
- Implemented multi-era reward and stake-pool delegation accounting, resolving ledger discrepancies across Cardano protocol eras through verification against Cardano DB-Sync.
- Built snapshot-based bootstrapping that starts state modules from historical chain points in about 15 minutes, saving developers hours of full-chain replay when testing era-specific behaviour.
- Implemented deterministic rollback and replay using bounded undo logs and reward journals, keeping ledger state consistent across modules during chain reorganizations.
- Helped build a purpose-built Cardano indexer for Midnight using Acropolis components, exposing cNIGHT and UTXO activity, governance and system transactions, validator-selection data, and stable chain state through gRPC as a lightweight alternative to its existing observation stack.
- Contributed to its proof-of-concept integration with the Midnight node, improving a critical query by up to 25x, helping nodes reach chain tip 3x faster, and reducing projected storage from roughly 700 GB to 100 GB.

Data Engineer

Sept. 2024 – Oct. 2025

BitBird

Remote

- Designed a protocol-aware pipeline that decoded tens to hundreds of thousands of daily transactions across major Cardano decentralized exchanges and NFT marketplaces.
- Partnered with the web team to build query models and APIs for token pricing, trading activity, ownership, and transaction history.

Lead Data Engineer

Dec. 2022 – Jan. 2024

Clir Renewables Inc.

Vancouver, BC

- Led four engineers building Go APIs and time-series pipelines for utility-scale solar, setting technical direction across ingestion, validation, modelling, and reporting.
- Saved approximately 150 analyst hours per month by extending a worker-based system to classify unit status and reconcile solar error codes.

Lead Developer, Platform

Oct. 2021 – Dec. 2022

Clir Renewables Inc.

Vancouver, BC

- Built customer-facing APIs and a queue-based export platform for large wind and solar datasets.
- Introduced Terraform-managed AWS infrastructure, CI/CD automation, and shared observability across production services.

Education

University of British Columbia

Vancouver, BC

B.Sc. Computer Science

2012 – 2017

Lighthouse Labs

Vancouver, BC

Web Development Bootcamp

June – Sept. 2017

Technical Skills

Languages: Rust, Go, TypeScript, Python, SQL, Compact

Frameworks & Libraries: Tokio, Cariatid, Tonic, Prost, Pallas, Criterion.rs

Blockchain & Protocols: Cardano, Midnight, DeFi, smart contracts, UTXO systems, gRPC, Protocol Buffers, CBOR, blockchain indexing, cross-chain infrastructure

Tools & Platforms: AWS, Docker, Terraform, Linux, PostgreSQL, Fjall, k6, Prometheus, Grafana, CloudWatch, CircleCI